The documentation and process conversion measures necessary to comply with this revision shall be completed by:
8 February 2018.

INCH-POUND

MIL-PRF-19500/608E
7 November 2018
SUPERSEDING
MIL-PRF-19500/608D
w/ AMENDMENT 1
8 March 2010

PERFORMANCE SPECIFICATION SHEET

# SEMICONDUCTOR DEVICE, DIODE, SILICON, SCHOTTKY, POWER RECTIFIER, COMMON CATHODE, COMMON ANODE, DOUBLER, TYPE 1N6660, CASE MOUNT PACKAGE, QUALITY LEVELS JAN, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

## 1. SCOPE

1.1 Scope. This specification covers the performance requirements for a silicon, schottky, power rectifier diodes. Four levels of product assurance are provided for each device type as specified in MIL-PRF-19500.

1.2 Package outline. The device package is case mount TO-254AA (isolated) in accordance with figure 1 for all device types.

1.3 Maximum ratings. Unless otherwise specified, $T_A$ = +25°C.

| Type | $V_R$ (1) | $V_{RWM}$ (1) | $I_O$ (1) (2) (3) $T_C$ = 100°C | $I_{FSM}$ (1) tp = 8.3 ms $T_C$ = +25°C | $R_{\theta JC}$ (1) (4) Die 1 | $R_{\theta JC}$ (1) (4) Die 2 | $T_{STG}$ & $T_J$ |
|---|---|---|---|---|---|---|---|
| | V | V | A dc | A(pk) | °C/W | °C/W | °C |
| 1N6660<br>1N6660CCT1 | 45 | 45 | 15 | 300 | 1.65 | 1.65 | –65 to +150 |
| 1N6660R<br>1N6660CAT1 | 45 | 45 | 15 | 300 | 2.8 | 2.8 | |
| 1N6660DT1 | 45 | 45 | 15 | 300 | 2.8 | 1.65 | |

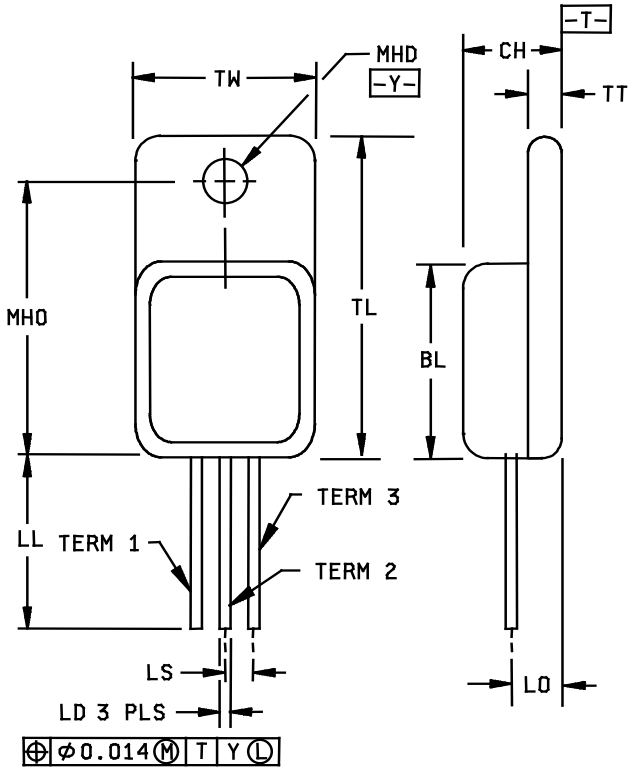
(1) Each individual diode.
(2) For derating see figures 2 and 3.
(3) Total package current is limited to 30 A dc.
(4) For thermal impedance graphs see figures 4 and 5.

1.4 Primary electrical characteristics. Unless otherwise specified, at $T_C$ = +25°C.
$R_{\theta JC}$ (entire package) = 0.85°C/W maximum for 1N6660CCT1.
$R_{\theta JC}$ (entire package) = 1.5°C/W maximum for 1N6660CAT1 and 1N6660DT1.
$R_{\theta JA}$ = 50°C/W maximum, each individual diode.

Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime, ATTN: VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

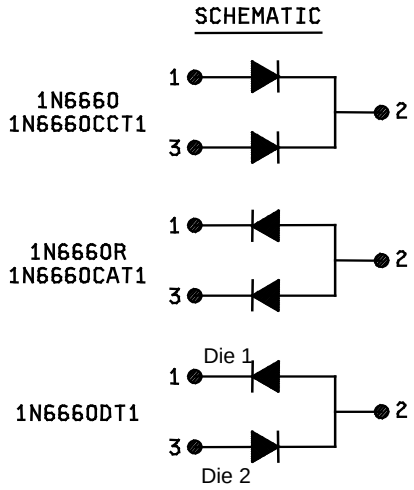


Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:03Z
Check the source to verify that this is the current version before use.




| Ltr | Dimensions | | | |
|---|---|---|---|---|
| | Inches | | Millimeters | |
| | Min | Max | Min | Max |
| BL | .535 | .545 | 13.59 | 13.84 |
| CH | .249 | .260 | 6.32 | 6.60 |
| LD | .035 | .045 | 0.89 | 1.14 |
| LL | .510 | .570 | 12.95 | 14.48 |
| LO | .150 BSC | | 3.81 BSC | |
| LS | .150 BSC | | 3.81 BSC | |
| MHD | .139 | .149 | 3.53 | 3.78 |
| MHO | .665 | .685 | 16.89 | 17.40 |
| TL | .790 | .800 | 20.07 | 20.32 |
| TT | .040 | .050 | 1.02 | 1.27 |
| TW | .535 | .545 | 13.59 | 13.84 |

NOTES:
1. Dimensions are in inches
2. Millimeters are given for general information only.
3. All terminals are isolated from case.
4. In accordance with ASME Y14.5M, diameters are equivalent to Φx symbology.




Types 1N6660, 1N6660CCT1, 1N6660R, 1N6660CAT1, and 1N6660DT1

FIGURE 1. Physical dimensions and configuration of TO-254AA package.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:03Z
Check the source to verify that this is the current version before use.

1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.4 for PIN construction example, 6.5 for a list of available PINs, and 6.6 for supersession information.

1.5.1 JAN certification mark and quality level. The quality level designators for encapsulated devices that are applicable for this specification sheet from the lowest to the highest level are as follows: "JAN", "JANTX", "JANTXV", and "JANS".

1.5.2 Device type. The designation system for the devices covered by this specification sheet is as follows.

1.5.2.1 First number and first letter symbols. The devices of this specification sheet use the first number and letter symbols "1N".

1.5.2.2 Second number symbols. The second number symbols for the devices covered by this specification sheet are "6660".

1.5.3 Suffix symbols. The following suffix symbol(s) are incorporated into the PINs for this specification sheet.

| | |
|---|---|
| | A blank designator identifies that the device is configured with a common cathode (see 6.6). |
| R | The "R" designator indicates that the device is configured with a common anode (see 6.6). |
| CAT1 | The "CAT1" designators indicates that the device is configured with a common anode. |
| CCT1 | The "CCT1" designators indicates that the device is configured with a common cathode. |
| DT1 | The "DT1" designators indicates that the device is configured as a doubler. |

1.5.4 Lead finish. The lead finishes applicable to this specification sheet are listed on QPDSIS 19500.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 – Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 – Test Methods For Semiconductor Devices.

(Copies of these documents are available online at https://quicksearch.dla.mil.)



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:03Z
Check the source to verify that this is the current version before use.

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturers list before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface requirements and physical dimensions. The interface requirements and physical dimensions shall be as specified in MIL-PRF-19500 and on figure 1 herein.

3.4.1 Lead finish, formation and material.

3.4.1.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein (see 6.6.3). Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4.1.2 Lead formation. When lead formation is performed, as a minimum, the vendor shall perform 100 percent hermetic seal in accordance with screen 14 of table E-IV of MIL-PRF-19500 and 100 percent dc testing in accordance with table I, subgroup 2 herein. Where a choice of lead formation is desired, it shall be specified in the acquisition document (see 6.2).

3.4.1.3 Lead material. The lead material shall consist of an iron-nickel or iron-nickel-cobalt alloy that is intended primarily for sealing the lead to the ceramic used for electrical isolation (see 6.6.3).

3.4.2 Electrical isolation of leads. Methods used for electrical isolation of the terminal feedthroughs shall employ materials that contain a minimum of 90 percent $Al_2O_3$ (ceramic).

3.4.3 Polarity. Polarity and terminal configuration of the device types shall be in accordance with figure 1.

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.7 Electrical test requirements. The electrical test requirements shall be as specified in tables I and II.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:03Z
Check the source to verify that this is the current version before use.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and tables I and II herein).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not require the performance of table III tests, the tests specified in table III herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

4.3 Screening (quality levels JANS, JANTXV, and JANTX only). Screening shall be in accordance with table E-IV of MIL-PRF-19500 and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

| Screen | Measurement | |
|---|---|---|
| | JANS level | JANTX and JANTXV levels |
| (1) (2) 3b | Surge current and impulse clamp voltage (see 4.3.1) | Surge current and impulse clamp voltage (see 4.3.1) |
| 3c | Thermal impedance (see 4.3.2) | Thermal impedance (see 4.3.2) |
| 3d | Avalanche energy test (see 4.3.3) | Avalanche energy test (see 4.3.3) |
| 4 | Required | Not applicable |
| 5, 8 | Required | Not applicable |
| 9, 10 | Not applicable | Not applicable |
| 11 | $V_{F1}$ and $I_{R1}$ | $V_{F1}$ and $I_{R1}$ |
| 12 | High temperature reverse bias (see 4.3.4) | High temperature reverse bias (see 4.3.4), t = 48 hours. |
| 13 | Subgroups 2 and 3 of table I herein, $V_{F1}$ and $I_{R1}$, excluding thermal impedance; $\Delta V_{F1}$ = ±50 mV (pk); $\Delta I_{R1}$ = ±25 µA dc or ±100 percent from the initial value, whichever is greater. | Subgroup 2 of table I herein, $V_{F1}$ and $I_{R1}$, excluding thermal impedance; $\Delta V_{F1}$ = ±50 mV (pk); $\Delta I_{R1}$ = ±25 µA dc or ±100 percent from the initial value, whichever is greater. |
| 15 | Required | Not applicable |

(1) Surge shall precede thermal impedance.
(2) Surge and thermal impedance shall be performed any time after sealing provided temperature cycling is performed in accordance with MIL-PRF-19500, screen 3a and before screen 13. JANTX and JANTXV levels do not need to be repeated in screening requirements.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:03Z
Check the source to verify that this is the current version before use.

4.3.1 Surge current and impulse clamp voltage. The surge current and impulse clamp voltage test shall be performed in accordance with condition A of method 4066 of MIL-STD-750. The following details shall apply: $I_O$ = 0, one current pulse, tp = 8.3ms, $I_{FSM}$ = 1.3 herein, and $V_{RWM}$ = 0.

4.3.2 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3101 or 4081 of MIL-STD-750 using the guidelines in that method for determining $I_M$, $I_H$, $t_H$, and $t_{MD}$. See table III, subgroup 4 herein.

4.3.3 Avalanche energy test. The avalanche energy test shall be performed in accordance with method 4064 of MIL-STD-750 using the circuit as shown on figure 6 or equivalent. The Schottky rectifier under test shall be capable of absorbing the reverse energy, as follows: $I_{AS}$ = 2 A minimum, L = 260 µH.

4.3.4 High temperature reverse bias. The high temperature reverse bias of devices shall be performed in accordance with test condition A of method 1038 of MIL-STD-750. The following details shall apply: $V_R$ = 36 V dc; $T_J$ = +125°C.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with table E-V of MIL-PRF-19500, and table I herein. Delta measurements shall be in accordance with the applicable steps of table II herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in tables E-VIA (JANS) and E-VIB (JAN, JANTX, and JANTXV) of MIL-PRF-19500 and as follows. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2, forward voltage test ($V_{F1}$) and reverse leakage test ($I_{R1}$) herein. Delta measurements shall be in accordance with table II herein.

4.4.2.1 Quality level JANS.

| Subgroup | Method | Condition |
|---|---|---|
| B4 | 1037 | $\Delta T_C$ = +85°C, $I_F$ = 2 A minimum for 2,000 cycles. |
| B5 | 1038 | Condition A, $V_R$ = 36 V dc, $T_J$ = +125°C, t = 340 hours min; heat sinking allowed. This test shall be extended to 1000 on each JANS wafer lot. |
| B6 | 4081 | 1N6660CAT1, 1N6660DT1 (die 1), $R_{\theta JC}$ = 2.8°C/W maximum for each leg.<br>1N6660CCT1, 1N6660DT1 (die 2), $R_{\theta JC}$ = 1.65°C/W maximum for each leg. |

4.4.2.2 Quality level JAN, JANTX, and JANTXV.

| Subgroup | Method | Condition |
|---|---|---|
| B3 | 1037 | $\Delta T_C$ = +85°C minimum, $I_F$ = 2 A minimum for 2,000 cycles. |



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:03Z
Check the source to verify that this is the current version before use.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500. Delta measurements shall be in accordance with table II herein.

| Subgroup | Method | Condition |
|---|---|---|
| C2 | 2036 | Condition A, weight = 10 lbs (4.54 kg), t = 15 seconds. |
| C6 | 1037 | $\Delta T_C$ = +85°C, $I_F$ = 2 A minimum for 6,000 cycles. |
| C6 | 1038 | Condition A, $V_R$ =36V dc, $T_J$ =125°C, t = 1000 hours minimum (for TX, TXV only); (heat sinking allowed). |

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the tests and conditions specified for subgroup testing in table E-IX of MIL-PRF-19500, and table III herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurements shall be as specified in section 4 of MIL-STD-750.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:03Z
Check the source to verify that this is the current version before use.

TABLE I. Group A inspection. 1/ 2/

| Inspection | MIL-STD-750 | | Symbol | Limit | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 1 | | | | | | |
| Visual and mechanical examination | 2071 | | | | | |
| Subgroup 2 | | | | | | |
| Thermal impedance 3/ | 3101 | See 4.3.2 | $Z_{\theta JC}$ | | | |
| 1N6660CAT1, 1N6660DT1 (die 1) | | | | | | °C/W |
| 1N6660CCT1, 1N6660DT1 (die 2) | | | | | | |
| Forward voltage | 4011 | $I_F$ = 5 A (pk) pulsed (see 4.5.1)<br>$I_F$ = 15 A (pk) pulsed (see 4.5.1)<br>$I_F$ = 30 A (pk) pulsed (see 4.5.1) | $V_{F1}$<br>$V_{F2}$<br>$V_{F3}$ | | 0.55<br>0.75<br>1.0 | V dc<br>V dc<br>V dc |
| Reverse current leakage | 4016 | Pulsed DC method (see 4.5.1), $V_R$ = 45 V dc, | $I_{R1}$ | | 1.0 | mA dc |
| Subgroup 3 | | | | | | |
| High temperature operation: | | $T_A$ = +125°C | | | | |
| Reverse current leakage | 4016 | Pulsed DC method (see 4.5.1), $V_R$ = 45 V dc | $I_{R2}$ | | 40 | mA dc |
| Low temperature operation: | | $T_A$ = –55°C | | | | |
| Forward voltage | 4011 | $I_F$ = 15 A (pk) pulsed (see 4.5.1) | $V_{F4}$ | | 0.8 | V dc |
| Subgroup 4 | | | | | | |
| Junction capacitance | 4001 | $V_R$ = 5 V dc; f = 1 MHz;<br>$V_{SIG}$ = 50 mV (p-p) (max) | $C_J$ | | 2,000 | pF |
| Subgroup 6 | | | | | | |
| Surge | 4066 | $I_{FSM}$ = 300 A (pk), 8.3 ms; pulsed (see 4.5.1) | | | | |
| Electrical measurements | | See table I, subgroup 2 herein | | | | |

See footnotes at end of table.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:03Z
Check the source to verify that this is the current version before use.

TABLE I. Group A inspection – Continued. 1/ 2/

| Inspection | MIL-STD-750 | | Symbol | Limit | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 7 | | | | | | |
| Dielectric withstanding voltage | 1016 | $V_R$ = 500 V dc; all leads shorted; V as measured from leads to case | DWV | | 10 | µA |
| Scope display evaluation | 4023 | Stable only | | | | |
| Electrical measurements | | See table I, subgroup 2 herein | | | | |

1/ For sampling plan, see MIL-PRF-19500.
2/ Each individual diode.
3/ For electrical measurements, this test is required for the following subgroups:
Group B, subgroups 3 and 4 (JANS).
Group B, subgroups 2 and 3 (JAN, JANTX, JANTXV).
Group C, subgroups 2, and 6.
Group E, subgroup 1.

TABLE II. Groups A, B, C, and E electrical end-point and delta measurements. 1/ 2/ 3/ 4/ 5/ 6/

| Step | Inspection | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|---|
| | | Method | Conditions | | Min | Max | |
| 1. | Forward voltage | 4011 | $I_F$ = 5 A (pk), pulsed (see 4.5.1) | $\Delta V_{F1}$ | ±50 mV dc from initial reading. | | |
| 2. | Reverse current | 4016 | Pulsed DC method (see 4.5.1), $V_R$ = 45 V dc, | $\Delta I_{R1}$ | ±25 µA dc or ±100 percent from initial reading, whichever is greater. | | |
| 3. | Thermal impedance<br>1N6660CAT1, 1N6660DT1 (die 1)<br>1N6660CCT1, 1N6660DT1 (die 2) | 3101 | See 4.3.2 | $Z_{\theta JC}$ | | | °C/W |

1/ Each individual diode.
2/ The electrical measurements for table E-VIA (JANS) of MIL-PRF-19500 are as follows:
a. Subgroup 3, see table II herein, steps 1 and 2.
b. Subgroup 4, see table II herein, steps 1 and 2.
c. Subgroup 5, see table II herein, steps 1 and 2.
3/ The electrical measurements for table E-VIB (JANTX and JANTXV) of MIL-PRF-19500 are as follows:
a. Subgroup 2, see table II herein, steps 1 and 2.
b. Subgroup 3, see table II herein, steps 1 and 2.
c. Subgroup 6, see table II herein, steps 1 and 2.
4/ The electrical measurements for table E-VII of MIL-PRF-19500 are as follows:
a. Subgroups 2 and 3, see table II herein, steps 1 and 2.
b. Subgroup 6, see table II herein, steps 1 and 2.
5/ The electrical measurements for table E-IX of MIL-PRF-19500 are as follows:
a. Subgroup 1, see table II herein, steps 1 and 2.
b. Subgroup 2, see table II herein, steps 1 and 2.
6/ Devices which exceed the table I, limits for this test shall not be accepted.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:03Z
Check the source to verify that this is the current version before use.

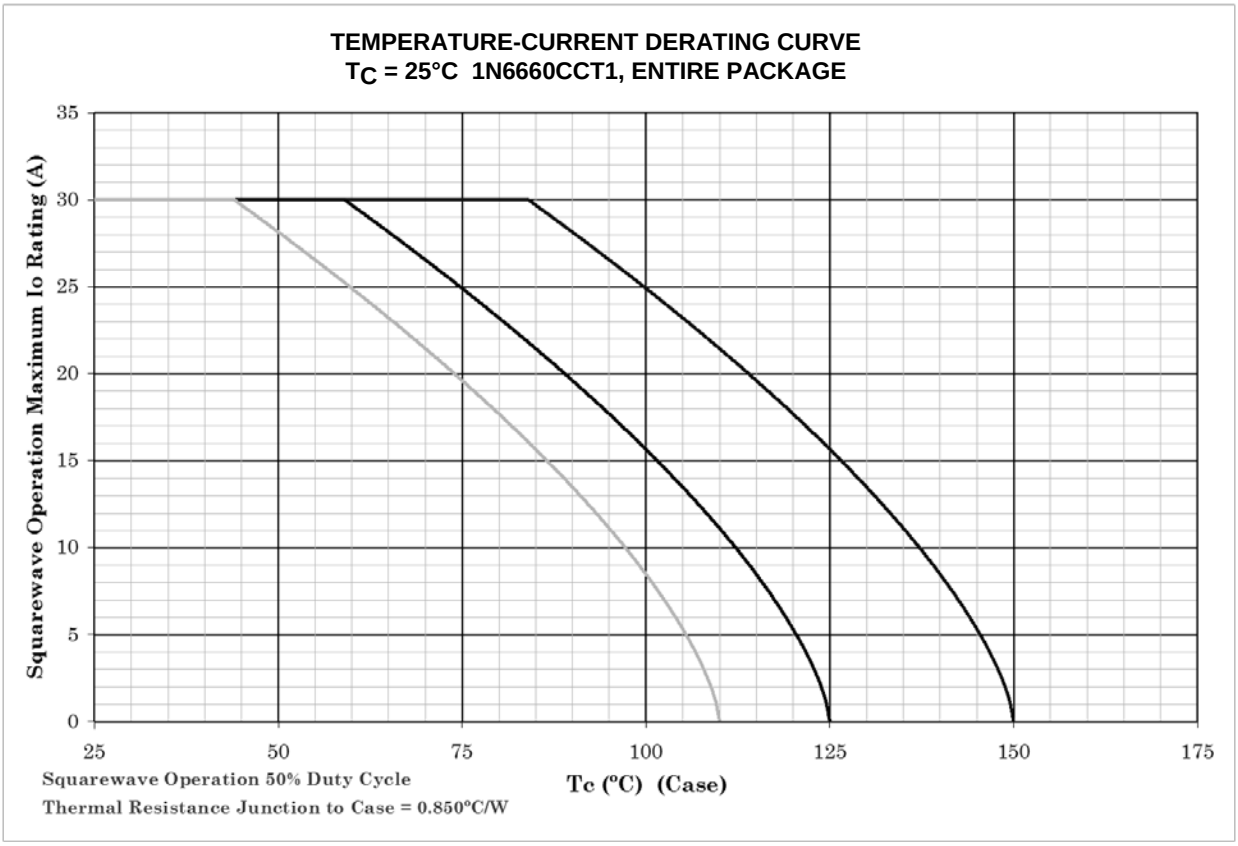
TABLE III. Group E inspection (all quality levels) for qualification and requalification only.

| Inspection | MIL-STD-750 | | Sample plan |
|---|---|---|---|
| | Method | Conditions | |
| Subgroup 1 | | | |
| Temperature cycling | 1051 | 500 cycles | n = 45, c = 0 |
| Hermetic seal | 1071 | | |
| Electrical measurements | | See group A, subgroup 2, and table II | |
| Subgroup 2 | | | |
| Steady-state blocking life | 1048 | t = 1,000 hours, $T_J$ = +125°C; $V_R$ = 36 V dc | n = 45, c = 0 |
| Electrical measurements | | See table I, subgroup 2, and table II | |
| Subgroup 3 | | | |
| Not applicable | | | |
| Subgroup 4 | | | |
| Thermal impedance curves | 3101 | See MIL-PRF-19500 | |
| Subgroup 6 | | | |
| ESD | 1020 | | |
| Subgroup 10 1/ | | | |
| Surge | 4066 | Condition A; $T_A$ = +25°C; $I_{FSM}$ = 300 A, 100 surges | n = 22, c = 0 |
| Electrical measurements | | See table I, subgroup 2 | |

1/ For each diode.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:03Z
Check the source to verify that this is the current version before use.



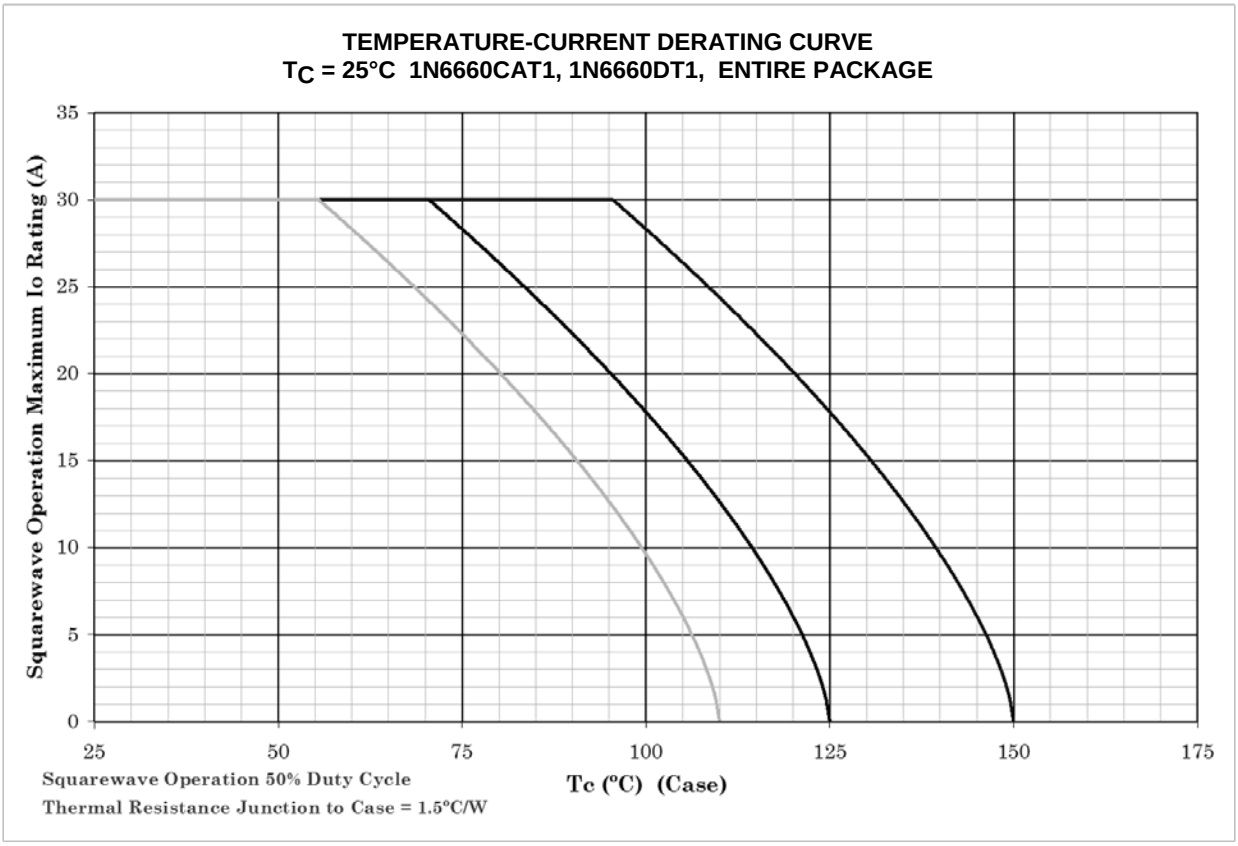

NOTES:

1. This is the true inverse of the worst case thermal resistance value. All devices are capable of operating at ≤ $T_J$ specified on this curve. Any parallel line to this curve will intersect the appropriate current for the desired maximum $T_J$ allowed.
2. Derate design curve constrained by the maximum junction temperature ($T_J$ ≤ 150°C) and power rating specified (see 1.3 herein.)
3. Derate design curves chosen at $T_J$ ≤ 125°C, and 110°C to show current rating where most users want to limit $T_J$ in their application.

FIGURE 2. Temperature-current derating graph.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:03Z
Check the source to verify that this is the current version before use.




NOTES:

1. This is the true inverse of the worst case thermal resistance value. All devices are capable of operating at ≤ $T_J$ specified on this curve. Any parallel line to this curve will intersect the appropriate current for the desired maximum $T_J$ allowed.
2. Derate design curve constrained by the maximum junction temperature ($T_J$ ≤ 150°C) and power rating specified (see 1.3 herein.)
3. Derate design curves chosen at $T_J$ ≤ 125°C, and 110°C to show current rating where most users want to limit $T_J$ in their application.

FIGURE 3. Temperature-current derating graph 1N6660CAT1, 1N6660DT1, die 1.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:03Z
Check the source to verify that this is the current version before use.

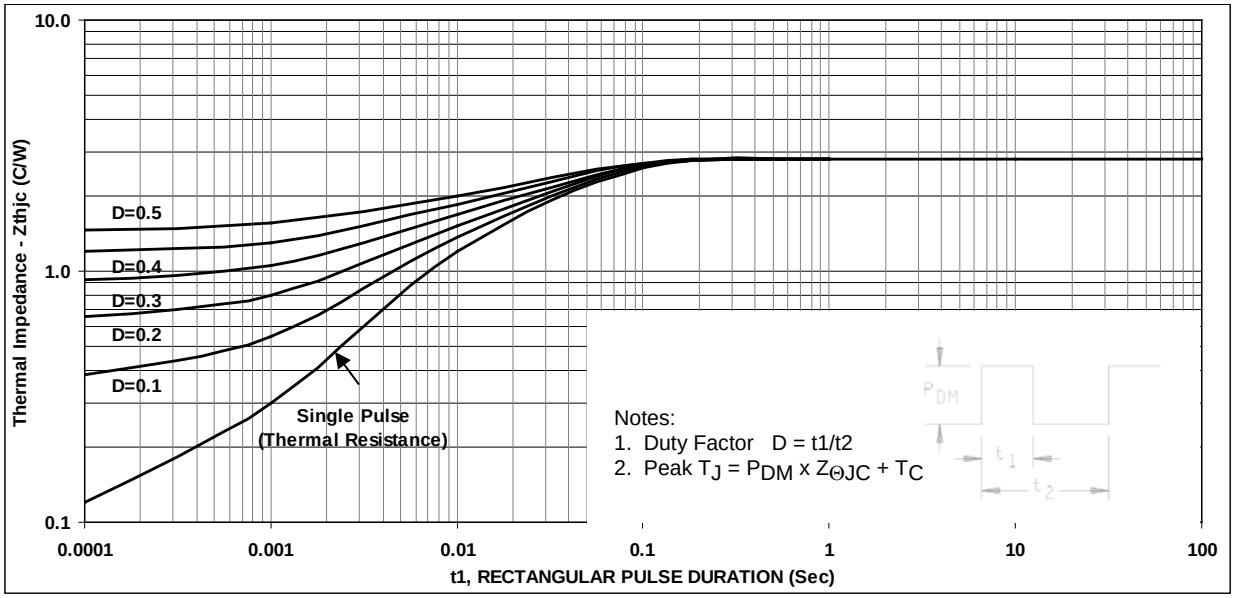



FIGURE 4. Thermal impedance for each leg 1N6660CAT1, 1N6660DT1, die 1, and 1N6660R.

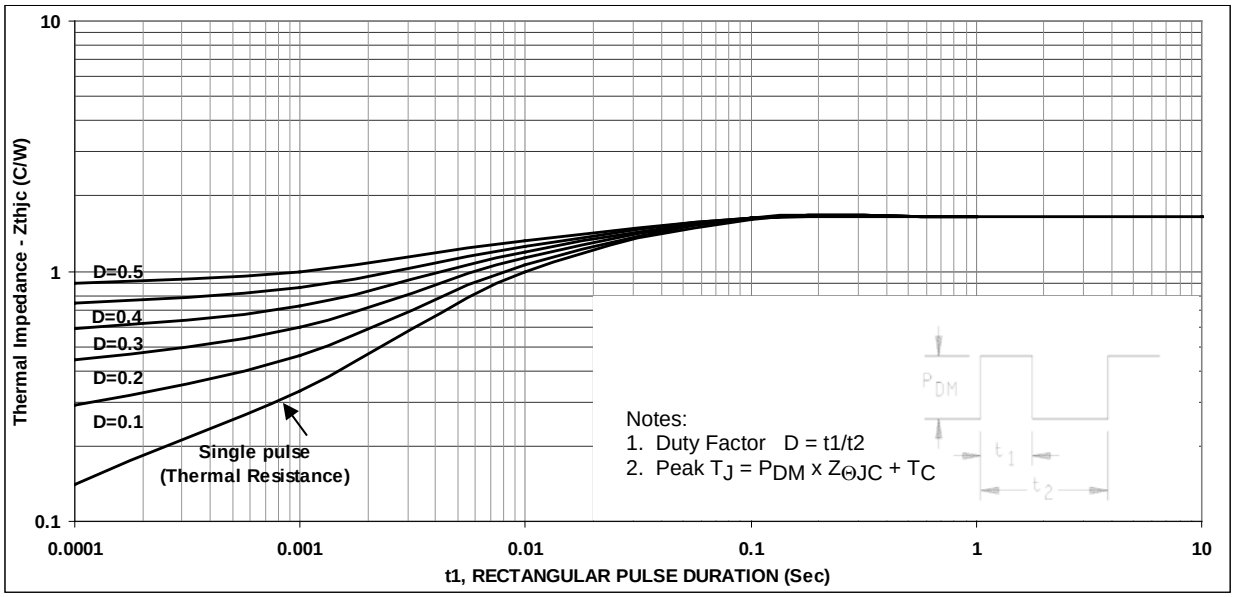
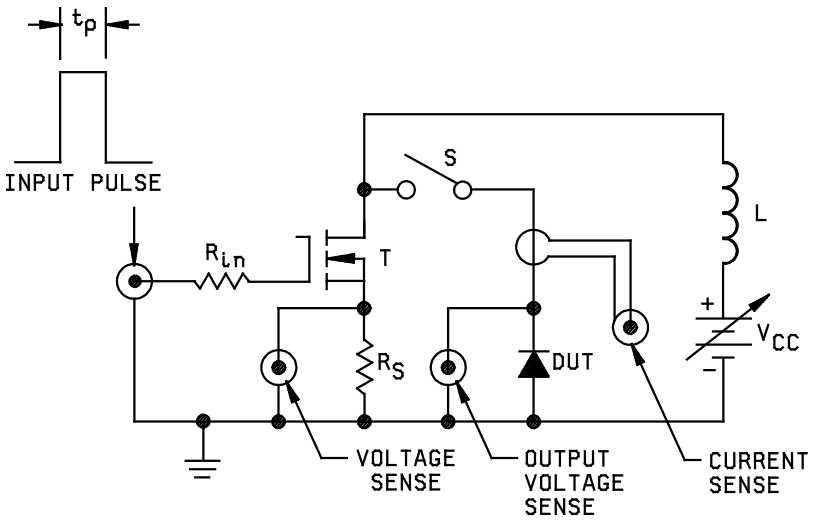



FIGURE 5. Thermal impedance for each leg 1N6660CCT1, 1N6660DT1, die 2, and 1N6660.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:03Z
Check the source to verify that this is the current version before use.




Input pulse: $R_{in}$ = 50 ohms
$V_G$ = 10 volts, $R_S$ = 0.1 ohms
$Z_G$ = 50 ohms, $V_{CC}$ ≈ 10 volts
L = 260 μH
Duty cycle ≤ 1 percent, T = 2N6756 or equivalent

PROCEDURES:

1. With S open, adjust pulse width to test current of 2 amps through $R_S$.
2. Close S, verify test current with current sense.
3. Read peak output voltage (see 4.3.3).

FIGURE 6. Avalanche energy test circuit.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:03Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

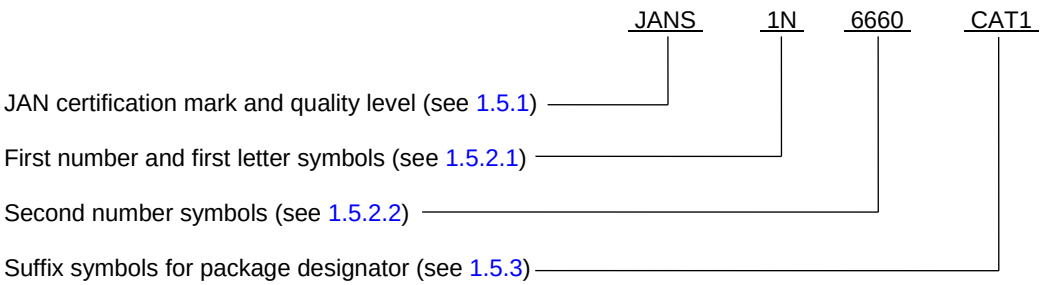
b. Packaging requirements (see 5.1).

c. Lead material, finish, and formation (see 3.4.1).

d. The complete PIN, see 1.5 and 6.4.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML-19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil . An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://qpldocs.dla.mil.

6.4 PIN construction example. The PINs for encapsulated devices are constructed using the following form.






Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:03Z
Check the source to verify that this is the current version before use.

6.5 List of PINs. The following is a list of possible PINs available for encapsulated devices covered by this specification sheet.

| JAN quality level | JANTX quality level | JANTXV quality level | JANS quality level |
|---|---|---|---|
| JAN1N6660CAT1 | JANTX1N6660CAT1 | JANTXV1N6660CAT1 | JANS1N6660CAT1 |
| JAN1N6660CCT1 | JANTX1N6660CCT1 | JANTXV1N6660CCT1 | JANS1N6660CCT1 |
| JAN1N6660DT1 | JANTX1N6660DT1 | JANTXV1N6660DT1 | JANS1N6660DT1 |

6.6 Replacement data.

6.6.1 Superseded PINs. The following supersession data applies to PINs associated with this document:

| Superseding PIN as specified within MIL-PRF-19500/608B, dated 17 August 2005 | Original PIN as specified within MIL-S-19500/608, dated 22 April 1993 |
|---|---|
| JAN1N6660CCT1 | JAN1N6660 |
| JAN1N6660CAT1 | JAN1N6660R |
| JANTX1N6660CCT1 | JANTX1N6660 |
| JANTX1N6660CAT1 | JANTX1N6660R |
| JANTXV1N6660CCT1 | JANTXV1N6660 |
| JANTXV1N6660CAT1 | JANTXV1N6660R |
| JANS1N6660CCT1 | JANS1N6660 |
| JANS1N6660CAT1 | JANS1N6660R |

6.6.2 Interchangeability. The PINs and device types 1N6660CCT1 and 1N6660CAT1 are interchangeable with the obsoleted PINs of device types 1N6660 and 1N6660R respectively.

6.6.3 Lead material. The original issue of this specification through MIL-PRF-19500/608A with Amendment 1 (22 February 1999) did not specify a lead material. MIL-PRF-19500/608B (17 August 2005) added a requirement for either "Alloy 52" or "Kovar" as the lead material. These lead materials were added to address issues with the TO-254AA package and to prevent latent package sealing problems. Because this document is classified as a performance specification sheet, this specific design solution for a lead material is no longer appropriate.

6.7 Request for new types and configurations. Requests for new device types or configurations for inclusions in this specification sheet should be submitted to: DLA Land and Maritime, ATTN: VAC, Post Office Box 3990, Columbus, OH 43218-3990 or by electronic mail at "Semiconductor@dla.mil" or by facsimile (614) 693-1642 or DSN 850-6939.

6.8 Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:03Z
Check the source to verify that this is the current version before use.

Custodians:
Army - CR
Navy - EC
Air Force - 85
NASA - NA
DLA - CC

Review activities:
Army - AR, MI, SM
Navy - AS, MC, OS
Air Force - 19

Preparing activity:
DLA - CC

(Project 5961-2018-081)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:03Z
Check the source to verify that this is the current version before use.